MUHANNAD YAZBAK

Full-Stack Developer & QA Automation Engineer
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PROFESSIONAL SUMMARY

Dynamic Software Engineer holding dual B.Sc. degrees, specializing in full-stack architecture, automated testing, and local/cloud system integrations. Proven track record building responsive Next.js applications, containerized AI microservices, and native desktop frameworks. Expert in implementing robust automation suites (Playwright/Selenium) via Page Object Model (POM) principles and optimizing continuous delivery pipelines.

TECHNICAL SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, Java, SQL
- **Web Frameworks & Tools:** React.js, Next.js, Node.js, Electron, Vite, Tailwind CSS
- **Automation & Testing:** Playwright (POM), Selenium, Appium, Vitest, Mockoon, E2E Testing
- **DevOps & Databases:** Docker, Git, GitHub Actions, CI/CD Pipeline Automation, SQLite, MySQL, MongoDB

CORE PROJECTS

TechMart – Full-Stack E-Commerce Ecosystem | *Next.js, TypeScript, Python, FastAPI, Docker, GitHub Actions, MySQL*

- **Architecture:** Engineered a decoupled, multi-cloud web platform separating an edge UI layer (Vercel) from containerized processing loops (Render) and an isolated relational registry (Aiven).
- **AI Integration:** Built an asynchronous Python RAG microservice with FastAPI and LangChain to process real-time semantic queries across Arabic, Hebrew, and English pipelines, mitigating LLM hallucinations.
- **Automation:** Authored a complete Playwright end-to-end testing framework applying the Page Object Model (POM) to validate client checkout funnels under simulated runtime scenarios.

Autonomous Playwright Self-Healing Test Agent | *Next.js, FastAPI, Python, Playwright, LLM APIs, GitHub Actions*

- **Core Engine:** Created an AI-driven testing ecosystem that dynamically intercepts broken locators (brittle CSS selectors or XPaths) at runtime, shipping broken DOM fragments to an LLM context layer for instant self-healing.
- **Code Generation:** Developed background processing scripts that autonomously observe automated human exploratory sequences on a target URL and output clean, structured, native spec.ts testing structures.
- **CI/CD Optimization:** Configured automated test runners inside GitHub Actions utilizing customized pip and npm dependency caching strategies to accelerate test execution speeds.

Localized Insurance Agency Management ERP | *Electron, Vite, React.js, TypeScript, Node.js, SQLite, Tailwind CSS*

- **Desktop Wrapper:** Assembled a secure, local-first CRM/ERP application utilizing Electron and SQLite to manage clients, active policy records, dashboards, and insurance claims with zero external data dependencies.
- **Local Ingestion:** Designed a custom drag-and-drop file module archiving sensitive physical media straight to the operating system's internal user storage paths using secure node fs handlers and localized database path mapping.
- **Disaster Recovery:** Integrated an upcoming background backup pipeline utilizing Google Drive OAuth2 synchronization to compress regional assets and automatically schedule encrypted database uploads daily.

EDUCATION & CERTIFICATIONS

- **2019 – 2023 | B.Sc. in Information Systems (BI Specialization)** – University of Haifa | *Focus on Python, SQL, ML, Big Data*
- **2010 – 2013 | B.Sc. in Medical Laboratory Sciences** – Ben-Gurion University of the Negev | *GPA: 88*
- **2023 | QA Automation Engineering Program** – Appleseeds (Elevation) | *Full E2E suite for Rami-Levy*
- **2021 – 2022 | Software Development Training** – Tsofen (Java, Spring Boot) & WebAhead (React, JS)